Circuit Rank versus Cubical β_1 in MindFold3D

The divergence analysis that motivated the topology-guaranteed β_1 generator: formal definitions, empirical characterization, and reproducibility artifacts

Preface

This document is part of the MindFold3D repository documentation and accompanies the article describing the MindFold3D procedural voxel-stimulus generator. It documents the mathematical relationship between two cyclic-structure descriptors: the circuit rank μ of the voxel adjacency graph — the quantity an earlier version of the generator controlled — and the first Betti number β_1 of the underlying continuum shape, the quantity the current framework targets and guarantees. The two are related but not identical, and their empirical divergence on generator output is what motivated the redesign to construction-guaranteed β_1 reported in the main manuscript. This appendix (i) gives formal definitions, (ii) shows why the two invariants differ in general — in both directions, (iii) reports an empirical cross-validation on a 240-shape corpus produced under the earlier μ-controlled design, (iv) documents the minimal diagnostic cases, and (v) provides code so every result can be reproduced from the public repository.

The main manuscript adopts cubical β_1 as the scored cyclic-structure feature, enforced on every emitted shape by template construction plus per-shape validation, precisely because the analysis below shows that μ is not a reliable proxy for volumetric holes on realistic generator output: the two agree on only about one-third of cyclic shapes under μ-controlled generation. Circuit rank remains in the production pipeline as an internal control quantity for the fill and optimizer loops — it is computable in linear time and stable under single-voxel perturbations — and is recorded on every shape as an unscored diagnostic. The topology_extras module implements the cubical β_1 computation used for validation and feature measurement.

A. Formal definitions

A.1 Circuit rank of the voxel adjacency graph

Let $S \subset \mathbb{Z}^3$ be a finite set of voxels (unit cubes with integer centers). The voxel adjacency graph $G(S) = (V, E)$ is defined by:

$$V = S$$
$$E = \{ \{u, v\} \subseteq S : u \text{ and } v \text{ share a face} \}$$

That is, two voxels are adjacent in $G(S)$ iff they share a 2D face (equivalently, 6-connectivity in the $\mathbb{Z}^3$ lattice). Let C denote the number of connected components of $G(S)$.

The circuit rank (also called the cyclomatic number or first Betti number of the graph G) is:

$$\mu(G) = |E| - |V| + C$$

$\mu(G)$ counts the number of independent cycles in G — equivalently, the minimum number of edges that must be removed to render G acyclic (i.e., to reduce G to a spanning forest) [Diestel, 2025, §1.9]. The identity $\mu(G) = |E| - |V| + C$ is a standard result: adding an edge either connects two components (increasing $|E|$ by 1, decreasing C by 1, leaving μ unchanged) or creates a new independent cycle (increasing $|E|$ by 1, leaving C unchanged, increasing μ by 1).

In the current MindFold3D pipeline, $\mu(G(S))$ is recorded as the unscored diagnostic circuit_rank; the scored topological feature betti_1 is the cubical β_1 of §A.2. Computation of μ is $O(|E| + |V|)$ via any standard adjacency enumeration.

A.2 First Betti number of the continuum shape

Given the voxel set S, one can construct a topological space $X(S) \subset \mathbb{R}^3$ by taking the union of closed unit cubes centered on each voxel. $X(S)$ is a cubical complex — a subcomplex of the standard cubical decomposition of $\mathbb{R}^3$.

The first Betti number $\beta_1(X(S))$ is the rank of the first singular (or simplicial, or cubical) homology group $H_1(X(S); \mathbb{Z})$. Intuitively, β_1 counts the number of independent 1-dimensional 'holes' in X(S) — features that a loop can wind around without contracting to a point. A solid torus has $\beta_1 = 1$; a figure-eight (as a 2-complex) has $\beta_1 = 2$; a solid ball has $\beta_1 = 0$.

For finite cubical complexes, β_1 is computable from the alternating-sum Euler characteristic and the Betti numbers β_0 (components) and β_2 (enclosed voids):

$$\chi(X) = \beta_0(X) - \beta_1(X) + \beta_2(X)$$
$$\beta_1(X) = \beta_0(X) + \beta_2(X) - \chi(X)$$

$\chi(X)$ is computable in linear time by counting cells (vertices, edges, squares, cubes) in the cubical decomposition. β_0 is the number of connected components of X (equivalently, of G(S), since two face-adjacent voxels are path-connected in X). β_2 is the number of enclosed voids, which can be computed by counting connected components of the complement $\mathbb{Z}^3 \setminus S$ under 6-connectivity and subtracting the unbounded component. Both β_0 and β_2 are $O(|E| + |V|)$.

The topology_extras module implements exactly this Euler-characteristic-based cubical β_1 computation, following the standard approach in digital topology (see, e.g., Bleile et al., 2022, for a careful treatment of cubical complex constructions and their homology).

B. Why circuit rank and cubical β_1 differ in general

The voxel adjacency graph G(S) and the cubical complex X(S) are two different mathematical objects derived from the same voxel set. G(S) is a 1-dimensional combinatorial structure; X(S) is a 3-dimensional topological space. They agree on β_0 (connected components) but not in general on higher Betti numbers.

The most direct way to see the discrepancy is to consider a 2×2 face patch: a flat square arrangement of four voxels in a single plane. This gives:

Property	Value	Explanation
|V|	4	four voxels
|E|	4	four face-adjacencies forming a square in G
C	1	one connected component
$\mu(G) = \|E\| - \|V\| + C$	1	the graph has one independent cycle (the square)
$\beta_1(X)$	0	the solid 2×2×1 slab is simply connected in $\mathbb{R}^3$

The graph has a cycle because there are four voxels arranged in a square with four face-adjacency edges. But the solid 2×2×1 slab of unit cubes has no 1-dimensional hole in the continuum: it is a contractible box. Any loop drawn in the solid slab can be shrunk to a point without leaving the slab.

More generally: circuit rank counts cycles in the 1-skeleton of the adjacency graph, while β_1 counts holes in the 3-dimensional realization. A cycle in the graph does not imply a hole in the shape whenever the cycle can be 'filled in' by a 2-cell in the cubical complex — as happens for any 2×2 face patch, and more generally for any locally 2-thick region of the shape. Conversely, narrow 1-voxel-thick tubes forming a loop will exhibit both $\mu(G) = 1$ and $\beta_1(X) = 1$.

The divergence also runs in the opposite direction. The adjacency graph sees only face-sharing: voxels that share only an edge or a corner are invisible to G(S), but their closed cubes still intersect in X(S). A configuration of edge-adjacent voxels can therefore enclose a genuine tunnel in the solid — $\beta_1(X) \geq 1$ — while the face-adjacency graph remains a tree ($\mu(G) = 0$). Section D.2 exhibits such a shape emitted by the production tree generator.

The two quantities are therefore equal when the shape is a 1-voxel-thick 'thin' structure with no edge-adjacency contacts between its strands, and can diverge otherwise, in both directions: $\mu(G)$ overcounts continuum holes wherever the shape contains 2×2 face patches (dense regions), and undercounts them wherever edge-adjacent strands enclose a tunnel.

C. Empirical characterization on a 240-shape corpus

To characterize the relationship between $\mu(G(S))$ and $\beta_1(X(S))$ on realistic generator output, we generated — under the earlier μ-controlled design that this analysis motivated replacing — a corpus of 240 shapes spanning all three archetypes (Tree, Chiral, Bridge) and all four Structural Complexity tiers (low, medium, high, expert), with 20 shapes per cell (3 archetypes × 4 tiers × 20 shapes = 240; see main manuscript §3 for tier definitions; corpus generation script: cross_validate_corpus.py).

For each shape, we computed both circuit_rank(G(S)) (the same code path used for the generator's internal control quantity) and cubical $\beta_1(X(S))$ (topology_extras). Corpus generation is deterministically seeded, so the archived per-shape data (cross_validation_per_shape.csv) and summary (cross_validation_results.json) regenerate byte-identically from the paper-v7 code. Overall agreement — i.e., $\mu(G) = \beta_1(X)$ — was observed in 83/240 shapes (34.6%). The fraction of shapes whose voxel set is a 1-voxel-thick 'thin' structure (containing no 2×2 face patches) was 85.8%.

C.1 Per-archetype-and-tier breakdown

Archetype/Tier	**n**	**Agree %**	**Thin %**	**μ range** *	*β_1 range**
tree/low	20	100	100	0–0	0–0
tree/medium	20	15	100	0–1	0–1
tree/high	20	0	100	2–2	0–1
tree/expert	20	0	85	4–5	0–1
chiral/low	20	100	100	0–0	0–0
chiral/medium	20	5	100	1–2	0–1
chiral/high	20	0	100	2–3	0–1
chiral/expert	20	0	45	4–5	0–0
bridge/low	20	100	100	1–1	1–1
bridge/medium	20	75	100	1–2	1–1
bridge/high	20	20	100	2–3	2–2
bridge/expert	20	0	0	12–12	0–0

C.2 Interpretation

The empirical results are consistent with the theoretical analysis in §B.

Low tiers (all archetypes): 100% agreement.

For the low-difficulty tier, MindFold3D produces small, mostly acyclic shapes with minimal geometric fill. Tree/low and Chiral/low shapes are single-branch chains ($\beta_1 = 0$), and Bridge/low produces a single loop through a 1-voxel-thick ring ($\beta_1 = 1$). In all cases the shape is thin, and $\mu(G) = \beta_1(X)$ by construction.

Tree and Chiral, medium-through-expert: near-total disagreement, in both directions.

Once fill voxels are added toward the target count, agreement collapses (0–15%). Most disagreements are overcounts: 2×2 face patches contribute graph cycles with no continuum hole, so μ reaches 5 while the skeleton remains tree-like. But the β_1 ranges of 0–1 in these cells also show the reverse direction inside the corpus itself: some nominally acyclic shapes contain one genuine tunnel enclosed by edge-adjacent strands (§D.2) that the face-adjacency graph cannot see. Neither invariant predicts the other in this regime.

Bridge/expert: extreme divergence (mean $\mu - \beta_1 = +12$).

The Bridge/expert tier deliberately produces geometrically dense shapes with many 2×2 face patches (thin fraction: 0/20). All 20 shapes exhibited $\mu(G) = 12$ and $\beta_1(X) = 0$: the tier requested four holes and delivered none, with the 12 graph cycles reflecting the face patches induced by dense packing. This is the empirical worst case in the corpus and, more than any other cell, the finding that made the μ-controlled design untenable for a hole-perception feature.

Overall implications.

$\mu(G)$ and $\beta_1(X)$ coincide for shapes lying in the 1-voxel-thick regime and diverge as soon as the shape acquires 2-thick local structure or edge-adjacency contacts — and the corpus exhibits both directions: overcounts dominate (2×2 face patches contributing graph cycles with no hole, up to $\mu = 12$ against $\beta_1 = 0$), while several nominally acyclic tree and chiral shapes contain a genuine edge-adjacency tunnel ($\beta_1 = 1$ with μ underneath it; §D.2). Overall agreement of 34.6%, with most μ-cyclic shapes containing no volumetric hole at all, is the central empirical finding: μ is a graph-connectivity quantity, not a hole count, and a difficulty feature whose cognitive rationale rests on hole perception cannot be operationalized through it. This finding motivated the current design, in which every cyclic shape is generated around templates that physically reserve its holes and every emitted shape is certified against a direct cubical β_1 computation (main manuscript, Topology-Guaranteed Generation).

D. Diagnostic case: the 2×2 face patch

The following minimal example is the fundamental unit of divergence between $\mu(G)$ and $\beta_1(X)$. It is worth stating explicitly because it identifies exactly which local configurations cause the two quantities to differ.

Configuration:

Four voxels arranged in a 2×2 planar square: $S = \{(0,0,0), (1,0,0), (0,1,0), (1,1,0)\}$.

Voxel adjacency graph G(S):

Vertices: $\{v_1, v_2, v_3, v_4\}$ (the four voxels)
Edges: $\{v_1v_2, v_1v_3, v_2v_4, v_3v_4\}$ (four face-adjacencies)
$|V| = 4, |E| = 4, C = 1$
$\mu(G) = |E| - |V| + C = 4 - 4 + 1 = 1$

Cubical complex X(S):

X(S) is the solid 2×2×1 slab of unit cubes. It is homeomorphic to a solid box and is contractible. By direct Euler-characteristic computation on the cubical decomposition, or by inspection, $\beta_1(X) = 0$.

Interpretation:

The four-cycle $v_1 \to v_2 \to v_4 \to v_3 \to v_1$ in G exists because each of the four voxels is face-adjacent to its two neighbors in the 2×2 arrangement. In the graph, this is a genuine cycle. In the cubical complex X, this cycle is the boundary of the 2-cell corresponding to the top face ($z = 0.5$) of the slab — it bounds a 2-disk and is therefore trivial in $H_1(X)$. This is the geometric origin of the overcount direction.

D.2 Diagnostic case in the opposite direction: the edge-adjacency tunnel

The reverse divergence — $\beta_1(X) > \mu(G)$ — occurs when strands of the shape touch only along cube edges, closing a tunnel in the solid that the face-adjacency graph cannot see. The following 15-voxel shape was emitted by the production TreeSkeleton pipeline (archetype tree, 7^3 grid):

$S = \{(0,3,2), (0,3,3), (1,3,3), (2,3,3), (3,2,3), (3,3,1), (3,3,2), (3,3,3),$
$(3,4,1), (3,4,3), (3,4,4), (3,5,2), (3,5,3), (4,3,3), (4,5,2)\}$

Its face-adjacency graph is a tree: $\mu(G) = 0$. Its cubical complex nevertheless contains one tunnel: $\beta_1(X) = 1$, verified both by the Euler-characteristic computation of §A.2 and by an independent GF(2) boundary-matrix homology computation (62 vertices, 123 edges, 76 faces, 15 cubes; rank $\partial_1 = 61$, rank $\partial_2 = 61$, giving $\beta_0 = 1$, $\beta_1 = 1$; cubical complexes embedded in $\mathbb{R}^3$ have torsion-free H_1, so the GF(2) rank equals the rational Betti number). The tunnel is enclosed by strands meeting along cube edges — contacts that contribute shared 1-cells to X(S) but no edges to G(S). This case is pinned as the regression test test_edge_adjacency_tunnel_beta1_exceeds_mu, and it is why acyclic tiers in the production pipeline validate $\beta_1 = 0$ explicitly rather than trusting the acyclicity of the adjacency graph: approximately 5–7% of 15-voxel tree shapes acquire such a tunnel.

E. Design consequence: construction-guaranteed β_1

The analysis above resolves the metric question for MindFold3D: the framework's cognitive rationale for cyclic structure rests on hole perception, so the scored feature must be $\beta_1(X)$, and §C shows $\mu(G)$ is not an acceptable proxy for it on realistic output. The obstacle — β_1's instability under single-voxel perturbation at 8–35-voxel scales (Johnson & Jung, 2021; Bleile et al., 2022) — is an obstacle to β_1 *as an incidental measurement*, not to β_1 as a construction target. The production generator therefore guarantees it: ring templates physically reserve each intended hole as inviolable empty space, fill and optimization cannot occupy the reserved regions or alter the certified topology (the optimizer re-verifies β_1 on every accepted swap), and a per-shape validation computes cubical β_1 directly, rejecting and regenerating any attempt that misses its target — including acyclic shapes, which are certified $\beta_1 = 0$ against the §D.2 failure mode. The main manuscript's Study 2 quantifies the cost of the guarantee: first-pass acceptance of 83–100% across the (β_1, voxel-count) space, at most 1.21 attempts per emitted shape.

$\mu(G)$ remains available on every emitted shape as the unscored diagnostic circuit_rank. It is the natural descriptor for applications in the graph-based voxel-object analysis tradition (Bhunre & Bhowmick, 2017; Ryan et al., 2025), and its relationship to β_1 (§C.1) documents the generator's fill density.

F. Reproducing the analysis

All code and data required to reproduce this appendix are included in the public MindFold3D repository at the release tag paper-v7, which corresponds to the exact code and result artifacts reported in the main manuscript. The relevant modules are:

mindfold3d/topology_extras.py — cubical β_1 computation.

Implements the Euler-characteristic method described in §A.2 (cubical_betti_1, with β_0, β_2, and χ also exposed). The accompanying test suite (tests/test_cycle_count.py, 12 tests) covers the edge cases, including the 2×2 face patch, solid cube and hollow shell (documented $\mu > \beta_1$ divergences), the §D.2 edge-adjacency tunnel ($\beta_1 > \mu$), and an optional cross-check against the gudhi library when installed.

prototypes/cross_validate_corpus.py — corpus generator and cross-validation script.

Regenerates the 240-shape corpus reported in §C under the earlier μ-controlled design (preserved in the prototypes directory for exactly this purpose) and recomputes both invariants per shape.

prototypes/investigate_divergence.py — diagnostic script.

Enumerates the specific graph cycles that contribute to $\mu(G)$ but not $\beta_1(X)$ and identifies the responsible local configurations.

studies/fidelity_study_hole_tiers.py and studies/benchmark_fidelity.py — the production studies.

Regenerate Study 2 (results/hole_tier_fidelity_N200.csv/.json) and Study 1 (results/benchmark_results_n200_allbetti.json) of the main manuscript against the production topology-guaranteed pipeline.

To reproduce all results in this document and the article:

```
From the root of this repository (optionally at the release tag
paper-v7.1 cited in the article):
pip install -r requirements.txt
python prototypes/cross_validate_corpus.py # regenerates the 240-shape corpus (§C)
python prototypes/investigate_divergence.py # regenerates the diagnostic table
python -m pytest tests/test_cycle_count.py # runs the 12-test unit suite (incl. §D.2)
python studies/benchmark_fidelity.py –n 200 –json results/benchmark_results_n200_allbetti.json
# Study 1 (deterministic, default seed 2026)
python studies/fidelity_study_hole_tiers.py # Study 2 (deterministic)
```

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